

SWETHA J

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate and AWS Certified AI Practitioner with expertise in Python, Java, Machine Learning, NLP, Generative AI, RAG, and Computer Vision. Experienced in developing AI-powered applications and intelligent automation solutions using LLMs and LangChain. Strong problem-solving, analytical, and software development skills with a focus on building scalable real-world AI solutions.

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science 2023 – 2027 | CGPA: 8.45
Honors in Cybersecurity and Data Privacy
RATHINAM TECHNICAL CAMPUS

SSLC | HSC 2023 | SSLC: 86.2 | HSC: 69.4
AMRITA VIDYALAYAM SENIOR SECONDARY SCHOOL - CBSE

TECHNICAL SKILLS

LANGUAGES / DATABASES: Python, Java, PostgreSQL
DATA SCIENCE / AI: Machine Learning, NLP, Computer Vision, Generative AI, LLM, Lang Chain, RAG, AI Agents, Chatbots, CNN, OpenCV, Prompt Engineering
LIBRARIES: Pandas, NumPy, Matplotlib, Seaborn, Streamlit
CLOUD & TOOLS: VS Code, Google Colab, AWS, APIs, Git, Linux, Excel, Power BI

STRENGTHS

Communication, Team Collaboration, Problem Solving, Critical Thinking, Time Management, Analytical Thinking, Adaptability

PROJECTS

Autonomous Smart Farming Agent

- Developed a full-stack AI-powered smart farming system for crop disease detection and real-time agricultural recommendations
- Integrated CNN, OpenCV, soil, and weather analysis along with an irrigation recommendation module for smarter decision-making
- Built scalable frontend, backend, and database workflows for real-time monitoring and prediction
- Tech Stack: React.js, Node.js, MongoDB, Python, OpenCV, CNN, APIs

Multi-Document Q&A System (RAG)

- Developed a full-stack AI-powered document question answering platform using RAG architecture and LLMs
- Implemented document upload, authentication, embeddings, and FAISS vector retrieval for efficient multi-document querying
- Generated accurate context-aware responses through LangChain pipelines and scalable backend integration
- Tech Stack: React.js, Node.js, MongoDB, Python, LangChain, Embeddings, Gemini API

Driver Drowsiness Detection System

- Developed a real-time driver monitoring system to detect fatigue and improve road safety
- Implemented facial landmark and Eye Aspect Ratio (EAR) techniques using computer vision
- Integrated alert mechanisms for immediate drowsiness detection and warning generation
- Tech Stack: Python, OpenCV, Dlib, NumPy, Computer Vision

AI Email Assistant Agent

- Built an AI-powered email assistant for email classification, summarization, and automated response generation
- Implemented LLM-based decision-making logic for smart prioritization and workflow automation
- Developed a real-time interface for efficient email management and response handling
- Tech Stack: Python, LLMs, NLP, Streamlit, APIs, Prompt Engineering

INTERNSHIP

UI/UX – Xortican Technologies

Designed user-friendly interfaces using wireframing and prototyping tools, focusing on improving user experience and usability.

Data Science & Analytics – hCue

Performed data preprocessing, visualization, and basic model building using Python libraries like Pandas, NumPy, and Matplotlib.

CERTIFICATES

- AWS AI PRACTITIONER
- AWS Machine Learning Foundations
- AWS Machine Learning for NLP
- Google Cybersecurity (Coursera)
- Using MongoDB with Java (MongoDB)
- Coding Essentials (Scalar)

LANGUAGES

- ENGLISH
- TAMIL
- HINDI